Trading Commodity Futures using Producer Equity Alpha

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1 Introduction

Commodity traders regularly model a wide variety of factors such as supply, demand, weather, and many others. Alternatively, the equity market attributes some α to companies whose sole or majority cash flow comes from commodities. These commodity producers have a direct relationship to commodities yet this link has not been fully well documented.
Below is a table of commodity futures that will be used with their respective start dates

Fut Ticker	Fut Name	date
BO	Soybean Oil	1980-01-03
SM	Soybean Meal	1980-01-03
SB	Sugar	1980-01-03
S	Soybean	1980-01-03
LC	Live Cattle	1980-01-03
W	Wheat	1980-01-03
KC	Coffee	1980-01-03
FC	Feeder Cattle	1980-01-03
CT	Cotton	1980-01-03
CC	Cocoa	1980-01-03
GC	Gold	1980-01-03
CL	Crude	1983-04-04
LH	Lean Hogs	1986-04-03
HO	Heating Oil	1986-07-08
CO	Brent Crude	1988-06-27
HG	Copper	1988-12-08
QS	Gas Oil	1989-07-05
NG	Natural Gas	1990-04-05
XB	Reformulated Gasoline	2005-10-05

With regards to equity alpha component, VanEck ETFs are used. The following ETFs are used.

ticker	ETF Name	Start Date
OIH	VanEck Oil Services ETF	2001-02-27
GDX	VanEckGold Miners ETF	2006-05-23
SMOG	VanEck Low Carbon Energy ETF	2007-05-10
MOO	VanEck Agribusiness ETF	2007-09-06
HAP	VanEck Natural Resources ETF	2008-11-12
CRAK	VanEck Oil Refiners ETF	2015-08-21
EMET	VanEck Copper and Green Metals ETF	2021-11-12

2 Base Model: ETFs

Start by building defining this relationship between commodity futures and ETFs that use those commodities which can be defined naively from the table below.

Name	Commodity Future
VanEck Agribusiness ETF	Cocoa, Coffee, Cotton, Feeder Cattle, Lean Hogs, Live Cattle, Soybean, Soybean Meal, Soybean Oil, Sugar, Wheat
VanEck Copper and Green Metals ETF	Copper, Nickel, Silver, Zinc, Carbon
VanEck Low Carbon Energy ETF	Carbon
VanEck Natural Resources ETF	Soybean Oil, Corn, Cocoa, Cotton, Gold, Copper, Heating Oil, Coffee, Aluminum, Live Cattle, Lead, Nickel, Tin, Zinc, Palladium, Platinum, Gas Oil, Soybean, Sugar, Silver, Soybean Meal, Wheat, Brent Crude, Feeder Cattle, Lean Hogs, Natural Gas, Reformulated Gasoline, Crude
VanEck Oil Refiners ETF	Gas Oil, Heating Oil, Reformulated Gasoline, Natural Gas
VanEck Oil Services ETF	Crude, Brent Crude
VanEck Gold Miners ETF	Gold

The setup of the model is built on two regressions. The general goal is to use the excess equity return of commodity producers to trade commodity futures. The first step towards separating the excess returns is running a rolling regression. In this case, take the rolling regression of the equity ETFs using S&P returns.

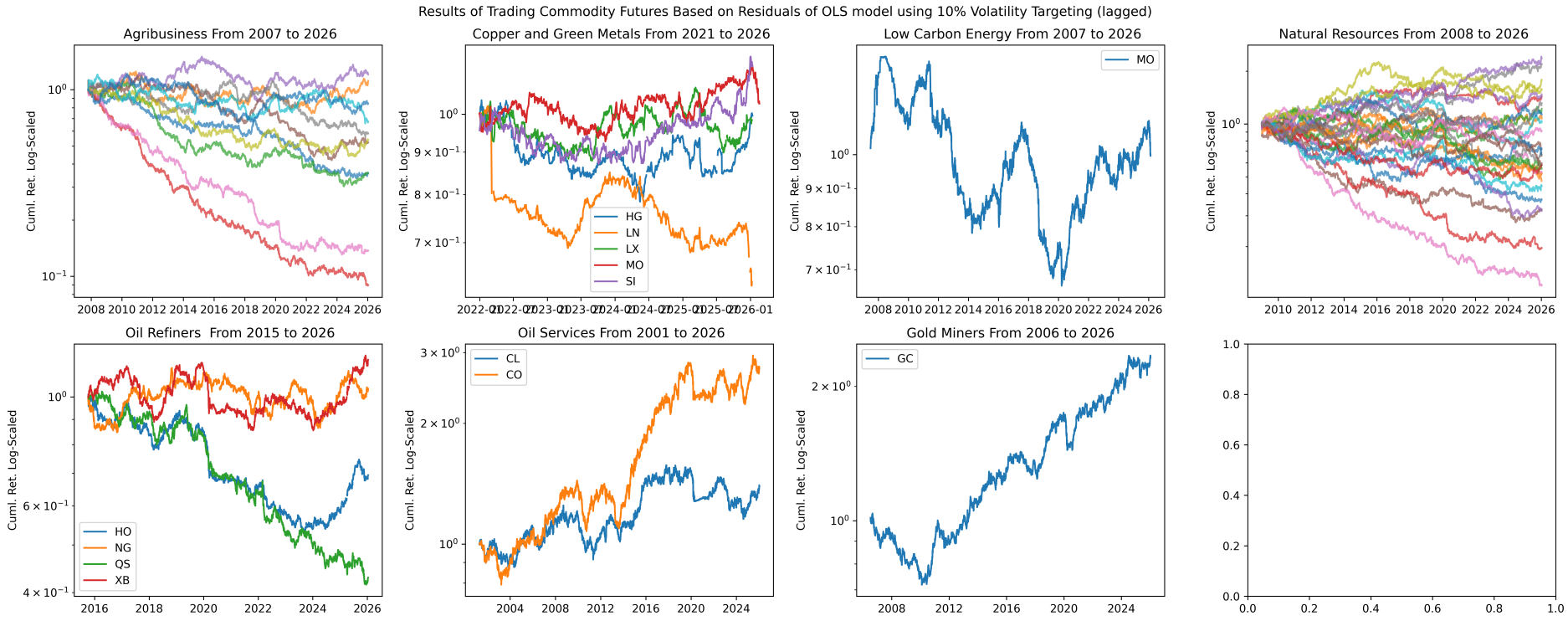
$$r_{t,\tau,ETF} = \beta_{t,\tau} \cdot r_{t,\tau,spx} + \alpha_{t,\tau} + \epsilon_t \tag{1}$$

For now, set τ which is the lookback window 30 days. The $\alpha_{t,\tau}$ is the time-varying returns that are not attributable to the market returns. The simplicity of the model is that, assuming the model's fidelity, the α is derived from the market, and therefore understanding the underlying structure is less relevant. This model essentially trades the *equity-market-derived* consensus for commodity futures.

Now plug that $\alpha_{t,\tau}$ (now referred as $\hat{\alpha}_{t,\tau}$) into the regression used for commodity futures. Which the form

$$r_{t,fut} = \beta_t \cdot \hat{\alpha}_{t,\tau} + \alpha_t + \epsilon_t \tag{2}$$

To ensure the relationship and ensure a causal linear relationship, the same regression as lagged $\hat{\alpha_{t,\tau}}$ is used. The regression results are quite large about 52 regressions, which leads to 104 when running with lagged parameters. Setting a p-value threshold of 10% and t-stat as $|1.6|$ the table below shows the number of regressions that resulted in statistically significant results.

Param	Lag	Stat. Sig.	Total	Ratio
α	Lagged	19	52	0.365
	No Lag	39	52	0.750
β	Lagged	18	52	0.346
	No Lag	21	52	0.404

Below are the results of trading the residuals of OLS controlling for different ETFs